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DETAIL SPECIFICATION

DETECTOR, COMBINED CONTAMINATED FUEL

This specification is approved for use by the Naval Air Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for two types of combined contaminated fuel detectors. This specification is not applicable to any +100 additized petroleum products.

1.2 Classification. The combined contaminated fuel detectors covered by this specification are classified as follows:

Type I	Motorized pump with visual free water detection capability (Type I will no longer be valid for procurement one year from publication date)
Type II	Motorized pump with digital direct-reading free water detection capability

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to: Commander, Naval Air Warfare Center Aircraft Division, Code 412000B120-3, Highway 547, Joint Base MDL, NJ 08733-5100 or emailed to michael.sikora@navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 6630

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DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-5624	Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-DTL-81248	Detector Pad, Free Water
MIL-S-81282	Standard, Free Water Detector

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-1399-300	Electric Power, Alternating Current
MS35058	Switch, Toggle, One Pole, Sealed Toggle
MS35059	Switch, Toggle, Two Poles, Sealed Toggle

(Copies of these documents are available online at http://quicksearch.dla.mil or https://assist.dla.mil.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM D3240	Standard Test Method for Undissolved Water in Aviation Fuels

(Copies of this document are available from www.astm.org.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA Publication 70	National Electrical Code

(Copies of this document are available from National Fire Protection Association (NFPA) at www.nfpa.org.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ANSI/NEMA HP 3	Electrical and Electronic Polytetrafluoroethylene Insulated High-Temperature Hook-Up Wire, Types ET (250 V), E (600 V), and EE (1,000 V)

(Copies of this document are available from National Electrical Manufacturers Association NEMA) at www.nema.org.)

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INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 8217	Petroleum Products – Fuels (Class F) Specifications of Marine Fuels

(Copies of this document are available from the International Organization for Standardization (ISO) at www.iso.org.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Materials. Detectors shall be made from such materials and by such processes as to ensure compliance with the requirements of this specification and the lightest practical weight.

3.2.1 Metals. Metals used in the manufacture of the detectors covered by this specification shall be resistant to corrosion due to fuels, salt spray or atmospheric conditions when in storage or during normal service life.

3.2.2 Fungus proof materials. Materials that are nutrients for fungi shall not be used in the detectors.

3.3 Construction. The combined contaminated fuel detector shall be a portable unit capable of determining the amount of particulate contamination in a fuel sample using light transmission between membrane filters, and free (undissolved) water by comparing the fluorescence of a uranine coated free water detector pad to a fluorescing standard. The combined contaminated fuel detector shall be modular in design and be repairable at the first or second echelon repair levels. The free water detector shall be capable of being completely removed from the unit for repair and replacement purposes. The detector shall be capable of translating particulate contamination results using 47 millimeter filters. The detector shall be capable of being hand-carried by personnel to a fueling site for the purpose of analyzing fuel for particulate contamination or water content. The detectors shall withstand the normal strains, jars, and any other conditions incident to service use. The detector shall be compatible with and safe for use with middle distillate fuels (JP-5, JP-8, Jet A, Jet A-1, F-24, F-76, F-75, DMA grade Marine Gas Oil).

3.3.1 Weight. The weight of the completely assembled detectors with attachments, components and case, shall be not greater than 27.0 kilograms.

3.3.2 Dimensions.

3.3.2.1 External dimensions. The maximum external dimensions of the detector shall be as

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follows:

Height	0.5 meter (19.7 inches)
Width	0.5 meter (19.7 inches)
Depth	0.3 meter (11.8 inches)

3.3.2.2 Space or enclosure maximum dimensions of detector operation. The detector shall be capable of being operated in a space or enclosure of the following maximum dimensions:

Height	0.8 meter (31.5 inches)
Width	0.7 meter (29 inches)
Depth	0.6 meter (23.6 inches)

3.3.2.3 External dimensions of the free water detection portion (type II only). The external dimensions of the free water detection portion shall be as follows:

Height	0.09 meter (3.5 inches)
Width	0.28 meter (11 inches)
Depth	0.11 meter (4.25 inches)

3.3.3 Drawer. The detector shall have a drawer, compartment or other means for storage of accessory items such as membrane filters, forceps, calibration filters, spare fuses and lamps. The storage means shall be readily accessible to operating personnel and shall not interfere with operation of the detector.

3.3.4 Filter receptacle/photocell housing. The filter receptacle tray shall be of sufficient size to accommodate 47 millimeter diameter membrane filters. The tray shall be of such design as to slide into a fixed photocell housing and shall not bind at any point between the open and closed positions in the photocell housing. A small cutout shall be provided on the top surface of the tray to facilitate placement and removal of the membrane filters. The photocell shall be a Vactec VTS-7070A or equivalent as approved by the procuring activity and shall be fixed securely in the housing. The photocell shall be hermetically sealed against moisture and contact with middle distillate fuel. A frosted glass, which equally diffuses light across its surface area, shall be mounted between the filter receptacle tray and incandescent light, and shall be sealed with an O-ring or other means to prevent fuel from leaking into the detector case.

3.3.5 Bottle receiver assembly. The bottle receiver assembly shall consist of a bottle receiver, membrane filter holder, one hole rubber stopper (size 8), and a ground wire with plug. The bottle receiver shall be the upper portion of the assembly properly sized so the neck of the sample bottle may be inserted when the bottle is inverted for the vacuum filtration operation. The membrane filter holder is the lower half of the assembly upon which the membrane filters are placed for filtration, and which shall mate with the mouth of the vacuum flask using a one-hole rubber stopper through which the lower tube of the filter holder is inserted. The top and bottom halves of the bottle receiver assembly shall fit together snugly with a twisting motion to hold the membrane filters during the filtration operation without leaking.

3.3.5.1 Type II only. The bottle receiver assembly shall be capable of holding both a 25 mm and 47 mm pad. This may require additional assemblies or adapters.

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3.3.6 Switches. Toggle type switches in accordance with MS35058 and/or MS35059 shall be provided to control the particulate and free water detection light sources, and to turn the vacuum pump on and off.

3.3.6.1 Particulate light switch. The particulate light switch shall be installed such that it must be in the "on" position before the pump switch can be activated.

3.3.6.2 Free water detector light switch (type I only). The free water detector light switch shall be a momentary toggle type switch with three positions: start (momentary), on, and off.

3.3.7 Free water test pad.

3.3.7.1 Ultraviolet light comparison (type I only). The type I detector shall be constructed so that a 47 millimeter diameter test pad can be viewed for comparison with the free water standards under ultraviolet light (3660π).

3.3.7.2 ASTM D3240 compliance (type II only). The type II detector shall be constructed to comply with all provisions of ASTM D3240 and shall report free water values digitally. The detector shall be designed to accommodate the required 25 mm test pad. All accessory items required to accommodate the 25 mm test pad shall be provided with the detector.

3.3.8 Vacuum receiving vessel. The vacuum receiving vessel shall be large enough to contain at least 1200 milliliters of filtered fuel at one time. The drain line of the receiver shall permit complete drainage of the receiving vessel. The receiving vessel shall be fitted with a high level detector which shall prevent the vessel from being overfilled and allowing fuel to reach the vacuum pump.

3.3.9 Electrical.

3.3.9.1 Power requirements. The detector shall be designed to operate on ungrounded 115 volts AC, 60 Hz and type I shipboard power as specified in MIL-STD-1399-300.

3.3.9.2 Wiring. The external and internal wiring shall be orderly and assembled in such a manner that individual wires may be easily traced. Connections between components shall be made using Molex type connectors to the maximum extent possible for ease of field replacement. Screw terminals and soldered joints are also acceptable. Crimp-on solderless connections shall not be used. The wire used shall be in accordance with NEMA HP 3.

3.3.9.3 Electrical components safety. Voltage and/or amperage regulating devices, transformers and capacitors shall be sealed with a fuel resistant material to prevent contact of fuel or vapors with internal components of the devices. All other electrical components and connections shall be insulated to protect against shock to personnel.

3.3.9.4 Motor. The motor shall be an explosion proof type with a current limiting device to prevent overheating in the event of pump malfunction. The motor shall comply with the NFPA Publication 70, Article 500 for Group D, Class I, Division 1 locations.

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3.3.9.5 Display meter. The display meter for particulate readings shall be capable of being read, without interpolation, in increments of 0.001 milliamperes between zero and one milliampere.

3.3.9.6 Power cord. The external power supply cord shall be a minimum 18-3 grounded cable, at least 5 feet long, with a male "U" ground attachment. The cord shall be capable of being removed from the cabinet for internal storage.

3.3.9.7 Circuit protection. Circuit protection shall consist of panel mounted fuses, which can be replaced from the exterior of the cabinet or mechanical type circuit breakers, which can be reset from the exterior of the cabinet.

3.3.9.8 Vacuum pump safety interlock. An electrical circuit shall be incorporated into the unit which shall shut off the vacuum pump and illuminate an indicator lamp labeled "vacuum flask overfill" on the detector's top panel whenever the high level detector in the vacuum receiving vessel is actuated by overfilling the vessel. This interlock circuit shall automatically deactivate whenever fuel is drained from the receiving vessel below the level of the overfill detector. An additional interlock shall ensure the drain valve is closed before restarting the vacuum pump.

3.3.10 Finish. Unless otherwise specified in the contract or order, painting and finishing of the detectors shall be in accordance with the supplier's standard practice, provided the following minimum criteria are met or exceeded: All surfaces to be painted shall have been cleaned so as to be free of all foreign matter detrimental to painting. At least one coat of primer and one top coat or equivalent shall be applied.

3.3.10.1 Color. Unless otherwise specified by the acquiring activity, the detectors shall be furnished in the manufacturer's standard color.

3.3.11 Accessory items.

3.3.11.1 General. All components necessary for filtration and measuring of transmitted light shall be provided with each detector. Compartments, holders, and/or a drawer shall be provided for storage of accessories, such as membrane filters, sampling bottle, squeeze bottle, bottle receiver assembly, forceps, tygon plastic drain tubing, calibration (Wratten) filters, spare fuses, and spare incandescent (or LED) and UV lamps.

3.3.11.2 Sample bottle. A sample bottle with a one liter capacity shall be provided. The bottle shall be constructed of a shatterproof material which shall be resistant to middle distillate fuels. The mouth of the bottle shall be capable of being inserted into the mouth of the vacuum receiver without leakage and shall also be provided with a screw type cap. Lines shall be marked on the outside of the bottle at the 200 milliliter, 500 milliliter, and 800 milliliter fill levels. Two small notches shall be filed in the top rim of the bottle or similar means shall be provided to allow air to enter the bottle during vacuum filtration.

3.3.11.3 Wash bottle. A wash bottle with a 0.5 liter capacity shall be provided. The bottle shall be constructed of a flexible shatterproof material which shall be resistant to middle distillate fuels. A fixed dispensing tube type cap shall be included with the wash bottle.

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3.3.11.4 Drain tubing. The drain tubing shall be middle distillate fuel compatible tygon type tubing or equivalent, sized to fit securely on the barbed drain fitting, and shall be at least 4 feet in length.

3.3.11.5 Miscellaneous. Each detector shall also be equipped with one pair of flat tipped forceps, 100 membrane filters with a 47 millimeter diameter and 0.65 micron pore size (Millipore Corporation catalog number DAWP 04700 or equivalent as approved by the procuring activity), 1 number 8 rubber stopper, 1 spare incandescent (or LED) lamp, 1 spare ultraviolet lamp, 1 spare frosted glass and 2 spare fuses (except when mechanical circuit breakers are employed).

3.3.11.5.1 Detector pads and standards (type I only). The detectors shall be equipped with 50 free water detector pads (47 mm) conforming to MIL-DTL-81248, and three free water detector standards conforming to MIL-S-81282.

3.3.11.5.2 Detector pads and standards (type II only). The detectors shall be equipped with 50 free water detector pads (25 mm) conforming to ASTM D3240, Test Pads section.

3.3.12 Vacuum pump. The vacuum pump shall be capable of producing a minimum vacuum of 25 inches of mercury, and the pump discharge shall be to the outside of the detector case.

3.3.13 Calibration filters. A set of calibrated wratten filters shall be provided with each detector. The filters shall be calibrated by the manufacturer in accordance with 4.4.7 and shall be assigned a number equivalent to a given contamination weight.

3.3.13.1 Wratten filters. Wratten filters shall be 47 millimeter diameter, neutral density (ND) filters, one each with ratings at ND-0.10 and ND-0.20. The filters shall be individually wrapped, and labeled as a set with the corresponding equivalent contaminant weight.

3.4 Performance.

3.4.1 Sample type. Devices shall meet all of the requirements of this specification when any middle distillate fuel is chosen as the sample. For the purpose of testing, the referee fuel to be used shall be JP-5 fuel conforming to MIL-DTL-5624.

3.4.2 Light source.

3.4.2.1 Light bulbs. The light bulbs shall be easily accessible for field replacement without the use of tools.

3.4.2.2 Light intensity. The intensity of the incandescent or LED light for particulate testing shall be mechanically or electrically adjustable from the exterior of the cabinet top, so as to provide a meter reading over the range of 0.550 to 0.800 milliampere.

3.4.2.3 Stability. After a warm-up period of three minutes, when set at a meter reading of 0.600, the meter reading shall not drift by more than 0.005 milliampere in a period of three minutes and shall not fluctuate by more than 0.002 milliampere.

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3.4.3 Filtration time. The filtration system of the detector shall be capable of filtering an 800 milliliter sample of the referee fuel through two 0.65 micron membrane filters in series in a maximum of two and one-half minutes using a minimum vacuum of 25 inches of mercury (see 4.4.5).

3.4.4 Free water standard.

3.4.4.1 Fluorescence decrease (type I only). There shall be no decrease in fluorescence at the edges of the standard (as a result of positioning of the free water standard) when the type I detector is tested as specified in 4.4.3.2.1 and 4.4.8.1. The standard shall conform to MIL-S-81282 and shall be government furnished (see 6.4).

3.4.4.2 ASTM D3240 calibrating standard compliance (type II only). Type II detectors shall comply with all provisions of ASTM D3240, Calibrating Standard Section when tested as specified in 4.4.3.2.2 and 4.4.8.2.

3.4.5 Viewer assembly (type I only). The viewer assembly on the detector shall have the capability of indicating water content of fuel samples in parts per million when tested in accordance with 4.4.8.1. The viewer shall indicate the water content by comparing the detector pad with the four fluorescing standards. If water droplets are present in the sample, they will disperse on the pad and cause the treated pads to fluoresce under an ultraviolet light.

3.5 Identification.

3.5.1 Identification of product. The switches, viewers, accessory slots and other functional external parts of the detectors shall be legibly and durably marked in accordance with MIL-STD-130.

3.6.1 Free water standards.

3.6.1.1 Free water standards conformance to MIL-S-81282 (type I only). A total of 3 sets of standards conforming to MIL-S-81282 shall be provided with each detector unit. The standards shall be included in the envelope. Instructions for attaching the standard to the unit shall be included.

3.6.1.2 Free water standards conformance to ASTM D3240 (type II only). Standards for type II shall comply with all provisions of ASTM D3240.

3.6.3 Calibration curves. Each detector shall have its own calibration curve. This curve shall show the relationship between the difference in the detector milliammeter reading and the actual amount of contaminant in the fuel over the entire scale, expressed in milligrams/liter (mg/L) or milligrams/gallon (mg/gal). The calibration curve shall be checked in accordance with 4.4.7.

3.6.4 Wratten filters. Two Kodak Wratten gelatin neutral density filters (ND-0.10 and ND-0.20) or equal shall be provided with each unit. All Wratten filters shall be calibrated by the detector manufacturer in accordance with 4.4.7 and shall be assigned a number equivalent to a given contamination level. The filters shall then be included with the detector. Note: The Wratten filters as

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purchased are square. They shall be cut to a 47 mm diameter circle in order to fit in the sample slide. A correctly sized filter is required to properly calibrate the light transmittance.

3.7 Workmanship. The detectors, including all parts and accessories, shall be fabricated and finished in a workmanlike manner and shall be free from blemishes, defects, burrs, sharp edges and incorrect dimensions and markings of parts and assemblies.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection of the detectors shall consist of all tests and examinations of 4.4.2 through 4.4.9 (see 6.2.1).

4.2.1 First article sample. The first article sample shall be one (l) complete unit for test. The first article sample shall be a representative production item. The sample shall be plainly identified by a securely attached durably marked tag containing the following information:

FIRST ARTICLE SAMPLE

DETECTOR, Combined Contaminated Fuel, Type I or Type II as applicable
Manufacturer's Designation or Number
Name of Manufacturer
Part Number
Submitted by (Name) (Date) for First Article Inspection in accordance with the requirements of MIL-DTL-22612D(AS) under authorization (reference letter authorizing test)

4.2.3 First article test location. The first article test shall be conducted at the Naval Air Warfare Center, Aircraft Division in Patuxent River, MD. The center shall be notified in writing of the contract award and estimated date of first article unit availability. The first article unit shall be packaged securely and shipped to:

Naval Air Warfare Center
Naval Air Systems Command, AIR 4.4.5.1
Bldg 2360, PSEF, Unit 4
22229 Elmer Road
Patuxent River, MD 20670-1534

4.3 Conformance inspection.

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4.3.1 Lot. A lot shall consist of all the detectors forming a part of one contract or order and submitted for acceptance at one time.

4.3.2 End item. Each detector shall be subjected to the applicable inspections and tests of 4.4.2 through 4. 4.9. Any defect or test failure shall be cause to reject the individual detectors.

4.4 Test methods.

4.4.1 Sample type. Devices shall meet all of the test methods specified in 4.4 when any middle distillate fuel is chosen as the sample. For the purpose of testing, the referee fuel to be used shall be JP-5 fuel conforming to MIL-DTL-5624.

4.4.2 Visual inspection. Each detector shall be examined internally and externally to determine conformance with table I and other requirements of this specification.

TABLE I. Visual inspections.

Defect
Unit not complete with all component parts.
Unit not free of cracks, tears, dents or distortion.
Unit not free of rocking when placed on a flat surface.
Unit not free of sharp edges or burrs.
Component parts not properly secured to housing.
Type I detector test pad holder not capable of being inserted into housing, or secured within housing, or removed from housing.
Line cord not adequately secured within housing.
Line cord not free of cuts, tears, or abrasions.
Blades of plug (Line cord) not properly aligned.
Type I detector test pad holder not capable of being held in closed position when viewer assembly is tipped to any angle.
Identification marking not complete, legible or permanent.
Finish not free of scratches, chips, or blisters.
Unit not free of dirt, grease, or foreign matter.

4.4.3 Electrical.

4.4.3.1 Light intensity. When the light intensity for particulate testing is adjusted to its highest value, a reading of 0.800 or higher shall be obtained on the display meter. When the adjustment is set to its lowest value, a reading of 0.550 milliampere or lower shall be obtained.

4.4.3.2 Free water detector.

4.4.3.2.1 Free water detector (type I only). The free water detector portion of the unit shall be tested by operating the on/off and start switch and observing proper starting of the ultraviolet bulb and illumination of the free water standards.

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4.4.3.2.2 Free water detector (type II only). The free water detector portion of the unit shall be tested by operating the on/off switch and observing illumination of digital screen.

4.4.3.3 Voltage/frequency response. Connect the detector power cord to a variable voltage/frequency power supply. Turn the detector on and allow to warm up for 3 minutes at 115 VAC and 60 Hz. Adjust the display meter to an initial reading of 0.600 milliampere. Vary the voltage from 92 to 138 volts AC and the frequency from 56.7 to 63.4 Hz and record the display meter readings at each voltage/frequency setting shown in Table II. The meter reading shall not drift by more than 0.005 milliampere or fluctuate by more than 0.002 milliampere from an initial setting of 0.600 milliampere.

TABLE II. Voltage/frequency test.

Voltage (volts AC)	Frequency (Hz)		
	56.7	60.0	63.4
92			
115			
138			

4.4.4 Vacuum test. Close the drain valve and insert a vacuum gage with rubber stopper into the vacuum flask mouth. Turn the pump switch on and record the maximum vacuum obtained. The maximum acceptable vacuum is -25 in/Hg.

4.4.5 Filtration time test. The vacuum pump shall be sufficient to complete the filtration process in a maximum time of two and one half minutes. The filtration process shall consist of passing 800 ml of clean referee fuel through two 0.65 micron Millipore or equal filters. Clean referee fuel is fuel that has been filtered through 0.65 micron Millipore filters. The flow system shall be thoroughly examined to assure no leakage of fuel (see 3.4.3).

4.4.6 Vacuum flask overfill. With the 800 ml of fuel from 4.4.5 still in the vacuum flask, and the vacuum pump running, continue to add fuel to the flask until the overfill protection system is activated. Check that the pump has shut down, with no fuel carried over into the pump, and that the indicator lamp on top of the unit is illuminated. Open the drain valve and drain the fuel from the flask. The pump shall not restart until all of the fuel is drained, and the drain valve returned to the closed position.

4.4.7 Calibration of particulate detector.

a. Turn the particulate light source switch to the “on” position and allow the light source to warm up for two to three minutes.

b. Thoroughly dry the filter receptacle before calibrating fuel, which may be on the receptacle surface, and could harm the Wratten filters.

c. Remove the silver foil-wrapped Wratten calibration filters from their envelopes and unwrap them, leaving the filters exposed in the opened wrapper. The filter should not be exposed any more than necessary to avoid coating the filter with dust. Forceps shall

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always be used to grasp the filters and the area of contract should be within ¼ inch of the outer edge of the filter to avoid damaging the effective surface.

d. Adjust the light intensity until a steady milliammeter reading of 0.600 is obtained.

e. Pick up one of the Wratten filters using the forceps (hold filters only within ¼ inch of outer edge), insert filter into the filter receptacle, slide the receptacle into the photo cell housing, and record the milliammeter reading. A small chip or crack in the edge of the filter shall not affect the reading unless it extends into the illuminated part of the filter, which can be noted, when the filter is in the filter receptacle.

f. With forceps, remove the filter from the sliding receptacle and replace in opened wrappings. Replace the sliding receptacle and note milliammeter reading. If the milliammeter reading is not 0.600, readjust light intensity and repeat e.

g. Repeat e and f using the second Wratten filter.

h. Unless duplicate readings are desired, the light switch should be turned to the "off" position and the Wratten filters wrapped and returned to their envelopes.

i. A calibration plot shall be made in which the ordinate shall represent the difference in milliammeter readings from 0.0 to 0.13 milliamp with divisions at every 0.005 milliamp. The abscissa shall represent the amount of contaminant in a fuel from 0.0 to 10.0 milligrams per liter with divisions at every 0.2 milligram per liter and also the equivalent milligrams per gallon with divisions at every 1.0 milligram per gallon.

j. To obtain the calibration point, subtract the lower of the two milliammeter readings of the Wratten filters from the higher reading. This difference (change in reading) is plotted on the calibration chart versus the weight of contaminant per liter given for the set of filters supplied. A second point is plotted at 0 milligram per liter and 0.01 milliamp change in light reading. Draw a straight line between the two points and extend the line to the edge of the chart. This line is the calibration line for the detector used.

4.4.8 Test of free water standards.

4.4.8.1 Free water standards (type I only). The standards shall be positioned in the box so that the forward edge shall be flush with the over-hanging edge of the standard retainer strip. The fluorescence shall be observed. In lieu of actual performance of such testing, the acceptance of the first article sample (see 4.2) shall be considered as complying with requirements of 3.4.3 and 3.4.4.1.

4.4.8.2 Free water standards (type II only). The free water detector standard shall be tested in accordance with ASTM D3240.

4.4.9 Shock test. The unit shall be closed and lifted from the left side using the handle to a height of 6 inches and dropped onto a hard surface. Repeat for a total of 3 drops. Conduct the same procedure, lifting from the right side using the handle. After all six drops have been conducted the

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unit shall still perform its intended functions.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The detectors covered by this specification are intended for use at Naval shore activities and aboard ships. The combined contaminated fuel detector is comprised of two portable instruments used to quickly determine the quantity of solid contamination and free water present in middle distillate fuels. The detector processes a sample of a fuel on site in usually three to four minutes, and establishes the level of contamination in the fuel. The level of contamination is measured by using the principle of differences in light transmission through membrane filters. For greater accuracy and to eliminate any fuel color effect, the two membrane filters are used in series. The first filter traps the solid contaminant, plus any fuel color effect. The second filter is subjected to only fuel color effects. The light transmission through the two filters is measured by a light sensitive cell system. Determining the difference between the amount of light transmitted through the contaminated and clean filters establishes the level of solid contamination in the fuel. The established level is evaluated with the calibration curve supplied with each detector to ascertain if the fuel is contaminated beyond required limits. All components necessary for filtration and measuring of transmitted light are incorporated into this one serviceable instrument. Also incorporated is a free water detector conforming to the requirements of ASTM D3240 or MIL-S-81282. A fuel sample is passed through a uranine-coated free water detector pad which is inserted in the pad holder; under the lamp illumination, free water absorbed by the pad fluoresces, and yields the fuel free water reading in parts per million. This product is known to produce invalid results with +100 additized fuels and therefore should not be used to determine particulate or free water content of +100 additized fuel. Due to the natural fluorescence of some diesel fuels, procedures for testing diesel fuels in these units may differ from aviation (kerosene) fuels.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Total quantity desired.

c. If first article is required (see 6.2.1).

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d. Warning and special markings when required.
e. Finish and color other than manufacturer's standard (see 3.3.10).
g. If technical manual is required (see 6.3 - 6.3.2).

6.2.1 First article. When a first article is required for inspection and approval (see 3.1, 4.2, 6.2, and 6.3), the contract should specify the following provisions for first article inspection. When a contractor is in continuous production of these contaminated fuel detectors from contract to contract, consideration should be given to waive the first article inspections. If inspection is required, indicate:

a. That the first article inspection should be conducted at the Naval Air Warfare Center (NAWC), with advance notification provided prior to shipping the first article test unit to NAWC.

b. That the approval of first article samples or the waiving of the first article inspection should not relieve the contractor of his obligation to fulfill all other requirements of the specification and contract.

6.3 Service materials and manuals. A service package consisting of materials and manuals, as described below, should be furnished in an envelope with each unit.

6.3.1 Technical manual. Each data package should contain two copies of an approved technical manual. The manual should include a list of component parts (including vendor part numbers where applicable), complete instructions for installation, operation, replacement and maintenance of the unit. As soon as practical, after the award of the contract, the supplier will furnish an electronic copy of the proposed technical manual for review by representatives of the contracting officer. The manuals should be reviewed, corrected and approved without any delay in delivery of the items. In addition, the supplier should furnish an electronic version of the approved technical manual to the contracting officer.

6.3.2 Previously approved manuals. When the supplier has furnished an approved manual for the item within the previous two years, the provision of 6.3.1 may be waived upon written request by the supplier to the contracting officer.

6.4 Government furnished parts. To each successful bidder, the government should furnish the following for each type I unit:

a. Three sets of standards conforming to MIL-S-81282
b. One-hundred free water detector pads conforming to MIL-DTL-81248

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodian:
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